PLASTIBYTES & HIVC // STRATEGIC HARDWARE DOSSIER

PETRA v2: The Autonomous AI STEM Educator

A Ruggedized Handheld AI Companion & Connected MCP Hardware Ecosystem for Africa and Emerging Markets — Inspiring Students from Primary Science to Deep-Tech Hardware Ventures.

2.0.0 (Executive Production Draft) Q2–Q3 2027 Commercial Pilot

Principal Systems, Industrial Design & Growth Team Buea, Cameroon // PlastiBytes & HIVC

01 / PEDAGOGICAL REACH

"Think Big": Learning Across All Educational Tiers

Petra v2 is not a toy; it is an adaptive intellectual partner. Through dynamic prompt adjustment and interchangeable physical sensor kits, Petra scales its pedagogy from 8-year-old primary school discoverers to 24-year-old startup engineers prototyping market-ready products.

K-5

Primary School

Curiosity & Nature

Interactive "Show & Tell" using the ring-light camera. Petra identifies leaves, rocks, and recyclable plastics. Gamified multiple-choice quizzes on the color screen.

6–12

Secondary & High

Electronics & Coding

Ohm's law schematics, breadboard continuity testing, and basic robotics. Students write code and observe real-world execution on the wireless rover chassis.

Univ

Higher Education

Embedded IoT & CAD

ESP32 firmware architecture, Fusion 360 parametric design, 3D printing tolerances, and remote agricultural sensor telemetry networks.

Pro

Startup Incubator

Hardware Founders

Mentorship Module 6: "Launch & Scale." Product costing, grant writing (Tony Elumelu, Mastercard Foundation), low-volume manufacturing, and DFM validation.

02 / HANDS-ON CREATION

Real-World 3D-Printed Projects Built with Petra

Petra connects the digital AI tutor to physical creation. Students use **PlastiBytes 100% recycled PET filament** (made locally from discarded water bottles) to 3D print real functional machinery guided by Petra's step-by-step display.



ROBOTICS & BIOMECHANICS

3D-Printed Bionic Robotic Gripper

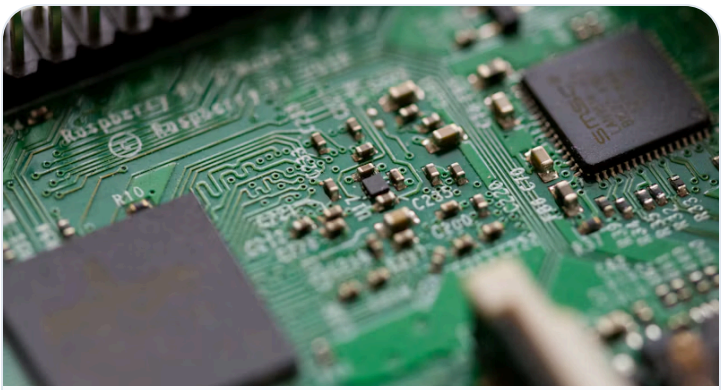
Students print articulate finger joints from recycled PET, wire servo motors to Petra's MCP expansion bus, and program touch reflexes.

3D Printing in Action

CLEAN WATER & IOT

Automated Water Tank Controller

Inspired by Ngum Love's university project: a 3D-printed weatherproof enclosure housing ultrasonic water sensors to prevent village tank overflows.



RENEWABLE ENERGY

Mini Wind Turbine & Solar Tracker

Aerodynamic 3D-printed turbine blades coupled with small DC dynamos. Petra monitors

power output and teaches real-time energy

♻️ Closing the Loop on African Plastic Waste

Over **125,000 plastic bottles** have already been collected and diverted from Cameroon's gutters by PlastiBytes. Petra demonstrates circular economics live: students shred water bottles, extrude filament, and print their own lab equipment.

03 / INDUSTRIAL DESIGN & TEARDOWN

Hardware Evolution: Prototype to Production



VERSION 1 — EXHIBITION BASELINE

Constructed for CITS26 Tech Summit. Modular breadboard wiring, fixed desktop orientation, 0.96" OLED with external ePaper panel.

- Seeed XIAO ESP32-S3 module
- Monochrome 128×64 mini screen
- Single omnidirectional PDM mic
- Fragile FDM 3D printed casing





— Handheld AI Unit, Smart Rover, Environmental Sensor Pod, Solar Panel & Recycled Filament.

MODULAR
PLUG & PLAY

04 / COMMERCIAL ARCHITECTURE

Business Model & Recurring Monetization

Rather than relying on one-off hardware sales, Petra v2 operates an integrated **Hardware + SaaS + Consumables ecosystem** with strong unit economics and predictable cash flow.

PILLAR 01 // HARDWARE

📦 Device & Kit Sales

Petra Solo: $79 USD (~49,000 FCFA). Handheld AI unit for individual students.

Petra Explorer Kit: $129 USD (~80,000 FCFA). Includes sensor pod and breadboard.

Classroom Lab Box: $1,499 USD. 10 handhelds, 2 rovers, charging hub, 10kg filament.

PILLAR 02 // SAAS SUBSCRIPTIONS

🎓 Mentorship & Software

Student Mentorship: $9.99/mo. 1-on-1 practitioner mentorship & project certification.

Teacher SaaS Portal: $199/yr per school. Automated grading, lab tracking, lesson builder.

Certification Exams: $25 fee for verified hardware engineering credentials.

PILLAR 03 // PHYSICAL CONSUMABLES

🧵 Recycled Filament Club

Filament Subscription: $35/mo for 2kg monthly recycled PET/PETG spools.

Direct Spool Sales: $18/spool (undercutting imported Chinese virgin filament by 30%).

Spare Sensor Packs: $15/pack for advanced sensor modules.

PILLAR 04 // INSTITUTIONAL

🏛️ CSR & B2G Grants

Telecom Subsidies: MTN & Orange sponsoring connectivity & devices for schools.

Govt Education Contracts: MINESEC (Cameroon), Rwanda Education Board.

Development Funds: Tony Elumelu Foundation, AfDB, UNDP, Mastercard Foundation.

3-Year Unit & Revenue Projections (Sub-Saharan Africa Pilot)

Metric / Revenue Stream	Year 1 (Pilot)	Year 2 (Regional Expansion)	Year 3 (Scale)
Hardware Units Shipped	1,200 units	8,500 units	35,000 units
Hardware Sales Revenue	$118,000	$840,000	$3,465,000
Mentorship & SaaS ARR	$36,000	$310,000	$1,540,000
Recycled Filament Consumables	$24,000	$185,000	$820,000
TOTAL REVENUE	$178,000	$1,335,000	$5,825,000
Blended Gross Margin	42%	51%	56%

05 / EXECUTION TIMELINE

Product Roadmap & Commercial Rollout

CITS26 Exhibition validation, prompt testing, and virtual smart-board deployment. COMPLETED / ACTIVE

Custom ESP32-P4+C6 PCB tape-out, dual beamforming acoustic tuning, and 10 hand-assembled functional units. ENGINEERING

VALIDATION

Injection mold tooling with recycled PET composite. 50-unit classroom trial across 5 Cameroonian secondary schools.

Initial 1,000-unit factory production run. B2G distribution and PlastiBytes Mentorship subscription expansion.

COMMERCIAL SCALE

PETRA v2 HARDWARE ARCHITECTURE & COMMERCIAL STRATEGY

© 2026 PlastiBytes & Hardware Innovation Valley Community (HIVC). All Rights Reserved.